

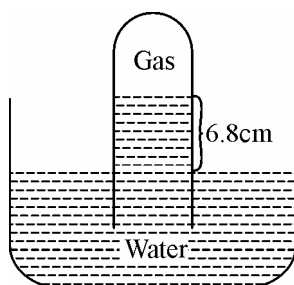
illuminati - 2019

Advanced Chemistry Assignment -1A | Physical Chemistry Class-Xlth

STOICHIOMETRY, ATOMIC STRUCTURE & GASEOUS STATE

SINGLE CHOICE QUESTION

- The average molecular speed is greatest in which of the following gas samples ?
 (A) 1.0 mol of N_2 at 560 K (B) 0.5 mol of Ne at 500 K
 (C) 0.20 mol of CO_2 at 440 K (D) 2.0 mol of He at 140 K
- A gas is collected by downward displacement of water at $T^\circ C$ according to the following diagram.



What is the value of $P_{\text{dry gas}}$ (in cm of Hg) ?

[Given : $d_{\text{Hg}} = 13.6 \text{ g/cm}^3$, $d_{\text{H}_2\text{O}} = 1 \text{ g/cm}^3$, vapour pressure of water $T^\circ C = 35$ torr, $P_{\text{atm}} = 76 \text{ cm Hg}$]

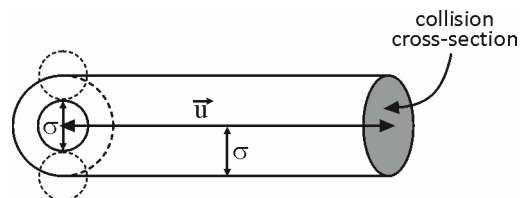
- (A) 68 (B) 72 (C) 75.5 (D) None of these
- At 298 K, the vapour pressure of water is 15.2 torr and relative humidity is 50%. The moles of water present in 2.446 litre of air is :
 (A) 10^{-3} (B) 0.10 (C) 2×10^{-3} (D) 0.20
- Photons of equal energy were allowed to strike on two different gas samples. One sample contains H-atoms in some-excited state with a principal quantum number 'n' and other sample has H-atoms in ground state. The photonic beams totally ionize the H-atoms in both samples. If the difference in the kinetic energy of the ejected electrons in the two different cases is 12.75 eV, then the principal quantum number 'n' of the excited state is :
 (A) 5 (B) 2 (C) 3 (D) 4
- A sample of ammonium sulphate was treated with 100 ml of 0.1 M NaOH and the resulting solution was boiled to expel the ammonia completely. It the excess base remaining in solution required 40 ml of 0.1 M HCl for complete neutralization then weight of $(NH_4)_2SO_4$ originally present in the solution is :
 (A) 0.66 g (B) 0.396 g (C) 0.792 g (D) None of these
- In a hypothetical galaxy, atomic numbers of all elements are the same as that on earth and Pauli's principle is obeyed. However, the shell closest to the nucleus is saturated with electrons before the next shell starts to fill and the process continues this way. Also, each orbital can accommodate only one electron having $+1/2$ spin. If noble element are defined as the elements with completely filled outermost shell, which element would be the noble element in their periodic table :
 (A) Ne (B) F (C) Si (D) Ar
- Titration of Na_3PO_4 with HCl to phenolphthalein end point produces Na_2HPO_4 , subsequent titration to methyl orange end point produces NaH_2PO_4 . Find the mass of Na_2HPO_4 present in a sample (containing Na_3PO_4 and Na_2HPO_4) which required 25 ml, 0.12 N HCl solution to reach the phenolphthalein end point and an additional 35mL of the same acid to reach the methyl orange end point.
 (A) $492 \times 10^{-3} \text{ g}$ (B) $170 \times 10^{-3} \text{ g}$ (C) $662 \times 10^{-3} \text{ g}$ (D) Data insufficient

8. If a water tank of 10^4 litre contains 55.5 g CaCl_2 , 47.5 g MgCl_2 , 84g, MgSO_4 and 81.6 g of CaSO_4 then what will be the degree of hardness in terms of ppm of CaCO_3 ? (Assuming density of solution in tank = 1 g/mL)
- (A) 23 ppm (B) 46 ppm (C) 130 ppm (D) 230 ppm
9. Bleaching powder (CaOCl_2) reacts with iodide ion as $\text{OCl}^- + 2\text{I}^- + 2\text{H}^+ \longrightarrow \text{I}_2 + \text{Cl}^- + \text{H}_2\text{O}$
 0.6 g sample of bleaching powder requires 34 ml of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ to titrate the liberated iodine. The % Cl in the sample is :
- (A) 10 (B) 20 (C) 30 (D) 40

Paragraph for Questions 10 - 11

Collision cross – section is an area of an imaginary sphere of radius σ around the molecule within which the centre of another molecule cannot penetrate. The volume swept by a single molecule in unit time is

$$V(\pi\sigma^2)\bar{u} \text{ where } \bar{u} \text{ is the average speed}$$

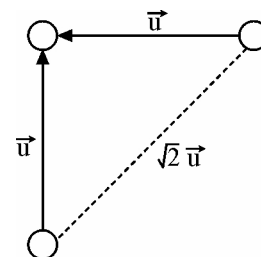


If N^* is the number of molecules per unit volume, then the number of molecules within the volume V is

$$N = VN^* = (\pi\sigma^2\bar{u})N^*$$

Hence, the number of collision made by a single molecule in unit time will be $Z = N = (\pi\sigma^2\bar{u})N^*$

In order to account for the movements of all molecules, we must consider the average velocity along the line of centres of two colliding molecules instead of the average velocity of a single molecule. If it is assumed that, on an average, molecules collide while approaching each other perpendicularly, then the average velocity along their centres is $\sqrt{2}\bar{u}$ as show below. Number of collision made by a single molecule with other molecules per unit time is given by



$$Z_1 = \pi\sigma^2(\bar{u}_{\text{rel}})N^* = \sqrt{2}\pi\sigma^2\bar{u}N^*$$

The total number of bimolecular collisions Z_{11} per unit volume per unit time is given by

$$Z_{11} = \frac{1}{2}(Z_1N^*) \text{ or } Z_{11} = \frac{1}{2}(\sqrt{2}\pi\sigma^2\bar{u}N^*)N^* = \frac{1}{\sqrt{2}}\pi\sigma^2\bar{u}N^{*2}$$

If the collision involve two unlike molecules then the number of collision Z_{12} per unit volume per unit time is given as

$$Z_{12} = \pi\sigma_{12}^2 \left(\sqrt{\frac{8kT}{\pi\mu}} \right) N_1N_2$$

When N_1 and N_2 are the number of molecules per unit volume of the two types of molecules σ_{12} is the average diameter of the two molecule and μ is the reduced mass. The mean free path is the average distance travelled by a molecule between two successive collisions. We can express it as follows :

$$\lambda = \frac{\text{Average distance travelled per unit time}}{\text{No. of collisions made by a single molecule per unit time}} = \frac{\bar{u}}{Z_1} \quad \text{or} \quad \lambda = \frac{\bar{u}}{\sqrt{2}\pi\sigma^2\bar{u}N^*} \Rightarrow \frac{1}{\sqrt{2}\pi\sigma^2N^*}$$

Two equal volume flasks containing equal masses of H_2 and CH_4 are at 100 K and 200 K. The molecular diameter of CH_4 is twice that of H_2 .

10. What is the correct relationship between Z_{11} of H_2 and Z_{11} of CH_4 ? [Z_{11} = number of bimolecular collisions per cm^2 per second.]
- (A) Z_{11} of $\text{H}_2 = 4 Z_{11}$ of CH_4 (B) Z_{11} of $\text{H}_2 = 8 Z_{11}$ of CH_4
 (C) Z_{11} of $\text{H}_2 = 16 Z_{11}$ of CH_4 (D) Z_{11} of $\text{H}_2 = 32 Z_{11}$ of CH_4
11. Select correct statement(s)
- (A) Mean free path of H_2 is twice that of CH_4
 (B) Number of collision per H_2 molecule per cm^3 per second is eight times that of CH_4
 (C) Average speed of H_2 is twice that of CH_4
 (D) Number of H_2 molecules per unit volume is four times that of CH_4 molecules.

Paragraph for Questions 12 - 14

When P_4 is heated in limited amount of O_2 , P_4O_6 is obtained and in excess of O_2 , P_4O_{10} is obtained .



12. What mass (in gm) of P_4O_{10} will be produced by the combustion of 32g of P_4 with 32g of O_2 ?
 (A) 8.32 (B) 16.64 (C) 32.09 (D) 48.64
13. How many moles of O_2 left unreacted in reaction (i) ?
 (A) 0.226 (B) 1.152 (C) 0.464 (D) 0.768
14. What mass (in gram) of P_4O_6 will be produced by the combustion of 32g of P_4 with 32g of O_2 ?
 (A) 15.84 (B) 31.9 (C) 60.16 (D) 78.24

MULTIPLE CHOICE QUESTIONS

15. Which of the following statement(s) is/are correct?
 (A) The ratio of circumference of third and second orbit of He^+ is 9 : 4
 (B) The ratio of spacing between the third and fourth orbit to the spacing between sixth and seventh orbit of H-atom is 7 : 13.
 (C) The ionization energy of Be^{3+} ion is x eV, then the ionization energy of He^+ ion should be $\left(\frac{x}{4}\right)$ eV
 (D) An electron revolving around H-nucleus in ground state absorbs 10.2 eV energy. Its angular momentum increases by $\left(\frac{h}{2\pi}\right)$.
16. 300 ml 0.05 M $Ca(MnO_4)_2$ in acidic medium can oxidize completely to :
 (A) 300 ml 0.5 M $FeSO_4$ (B) 50 ml 1 M FeC_2O_4
 (C) 150 ml 0.5 M $K_2C_2O_4$ (D) None of these
17. Commercial HCl is prepared by heating NaCl with H_2SO_4 according to following reaction.

$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

 Which of the following statement(s) is/are correct ?
 (A) 196 g of 80% of pure H_2SO_4 is required for production of 245 g of conc. HCl containing 40% HCl by weight.
 (B) 245 g of 80% H_2SO_4 by weight is required for the production of 365 g of conc. HCl containing 40% HCl by weight.
 (C) 2 mol of H_2SO_4 is required for the production of 365 g of 40% HCl by weight.
 (D) 468 gm of 50% NaCl by weight is required for the production of 365 g of 40% HCl by weight
18. Which of the following statement(s) is/are correct?
 (A) The vander waal's constant a is a measure of attractive force.
 (B) The vander waal's constant b is also called co-volume or excluded volume
 (C) Vander waal's constant b is expressed in $L \text{ mol}^{-1}$
 (D) Vander waal's constant b is one – third of critical volume.
19. 500 ml 22.4 V H_2O_2 solution and 500 ml 44.8 V H_2O_2 solution mixed together.
 Which is/are correct statement about final H_2O_2 solution if density equal to 1.182 gm/ml.
 (A) volume strength is 33.6 V (B) Molarity is 3 M
 (C) % w/v of H_2O_2 is 10.2 (D) Mole fraction of H_2O_2 is 0.0476
20. Ammonia (NH_3) gas combines with oxygen gas over Pt catalyst to produce Nitric oxide (NO) and water. If 6.8 g of NH_3 gas is taken initially then :
 (A) Volume of oxygen gas required at 1 atm and 273 K is 11.2 litre
 (B) Volume of $H_2O(l)$ produced at $4^\circ C$ (assuming density of water as 1000 Kg/m^3) is 21.6 ml.
 (C) Total mass of product obtained is 22.8 g
 (D) Number of moles of NO produced is 0.4

21. For two gases A and B with molecular weight M_A and M_B . It is observed that at a certain temperature T, the average speed of A is equal to root mean square speed of B. Thus the average speed of A can be made equal to the average speed of B.

Statement I : By keeping A at higher temperature than B.

Statement II : By keeping B at higher temperature

Statement III : By keeping A at $\left(\frac{8}{3\pi}\right)$ times temperature of B

Correct statement is/are :

- (A) Only Statement I is correct (B) Only Statement II is correct
(C) Only Statement III is correct (D) Statements I, II and III are correct

MATRIX MATCH TYPE

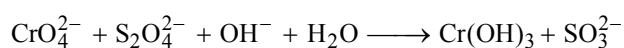
22. **Column I (orbitals)** **Column II (Nodal properties)**
- (A) 3s (P) Angular Node = 0
(B) 2p (Q) Radial node = 0
(C) 1s (R) Angular node = 1
(D) 3p (S) Radial node = 1

23. **Column I (orbital)** **Column II (Set of n, ℓ , m)**
- (A) 2s (P) $n = 4, \ell = 2, m = 0$
(B) $2p_z$ (Q) $n = 4, \ell = 1, m = +1$
(C) $4d_{z^2}$ (R) $n = 2, \ell = 1, m = 0$
(D) $4p_x$ (S) $n = 2, \ell = 0, m = 0$

24. **Column I** **Column II**
- (A) Repulsive forces dominate (P) $Z = \frac{3}{8}$
(B) At Boyle's temperature in the high pressure region for a gas (Q) $Z < 1$
(C) At very low pressure and at very high temperature (R) $Z > 1$
(D) At the critical point. (S) $Z = 1$

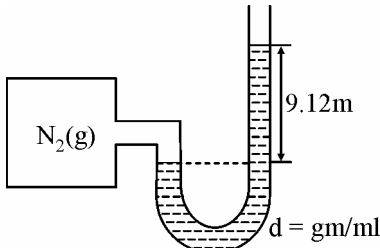
SUBJECTIVE QUESTIONS

25. In a chrome plating plant CrO_4^{2-} (Chromate) ions are present in waste water. The chromate ions are reduced to insoluble chromium hydroxide, $\text{Cr}(\text{OH})_3$, by dithionate ion $\text{S}_2\text{O}_4^{2-}$ in basic medium.



10 L of water requires 522 g of $\text{Na}_2\text{S}_2\text{O}_4$. If the normality of CrO_4^{2-} in waste water is $\left(\frac{x}{10}\right)\text{N}$. What is the value of x?

26. 5 mol of an equimolar mixture of CuS and Cu_2S will require x mol of KMnO_4 in acidic medium on complete oxidation. Calculate the value of x.
- 27.. During the excitation of an electron from ground state, it travels a distance of nearly 0.795 nm in the hydrogen atom. Calculate the maximum number of spectral lines formed during the de-excitation.
(Given : Radius of H atom in ground state = 0.53 Å)

28. Ca^{2+} responsible for hardness of water, and they be removed by passing hard water through a column of cation exchange resin (RH). When 500 ml hard water is passed through resin (RH) column. The water obtained needed 60 ml of 0.3 N NaOH for complete neutralization. Calculate the value of Z. $Z = \frac{1}{360} \times \text{Hardness of water}$.
29. A bulb of 8.21 litre volume filled with N_2 at 200 K is connected to glycerin manometer. The manometer shows 9.12 m difference in level. Calculate mol of N_2 present in bulb. (Given : $d_{\text{glycerin}} = 3.4 \text{ gm/ml}$; $d_{\text{Hg}} = 13.6 \text{ gm/ml}$, atmospheric pressure = 1 atm)
- 
30. A photon is emitted in β -line of Balmer series of He^+ ion. This photon is absorbed by an electron present in ground state of hydrogen atom. Calculate the value of z. Where $z = \frac{\lambda}{0.529\pi}$.
 λ = de-Broglie wave length associated with an electron in final orbit of Hydrogen atom. (in Å)
31. An electron is excited by a potential of 1.5 V from rest. If uncertainty in position is in order of de-Broglie wave length. Calculate the value of z.
 $z = 2\pi\Delta\lambda$ (λ in Å)
32. An unipositive charged particle accelerated from rest by a potential difference of 40 V is projected towards nucleus of a metal. What is the value of $\left(\frac{z}{10}\right)$?
 Given : z = atomic number of metal, closest distance of approach = 14.4 Å.

THERMODYNAMICS AND THERMOCHEMISTRY

SINGLE CHOICE QUESTION

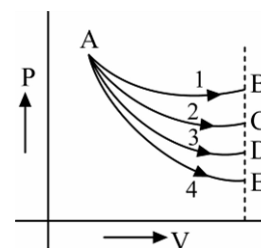
33. **Statement : I** The amount of work done in the reversible isothermal expansion is greater than work in the adiabatic system for the same final volume.
- Statement : II** In the adiabatic expansion of a gas temperature and pressure both decreases due to decrease in internal energy of the system.
- (A) Statement-I is True, Statement-II is True and Statement-II is a correct explanation for Statement-I.
- (B) Statement-I is True, Statement-II is True and Statement-II is NOT a correct explanation for Statement-I.
- (C) Statement-I is True, Statement-II is False.
- (D) Statement-I is False, Statement-II is True.

Paragraph for Questions 34 - 36

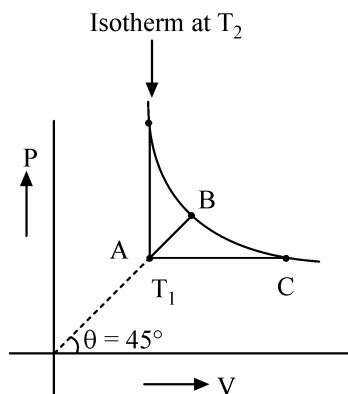
Molar heat capacity is the heat required to raise the temperature of one mole of a substance by 1 K. Heat required to change temperature in a process depends upon path followed.

34. Which gas has maximum value of C_{pm} ? (Assume vibratory motion active)
 (A) C_2H_2 (B) CO_2 (C) SO_2 (D) NH_3
35. Given AB, AC, AD and AE are reversible adiabatic processes for O_2 , CO_2 , SO_2 and NH_3 (may not be in same order). Assume vibratory degrees of freedom are active. Which is correct match ?

	O_2	CO_2	SO_2	NH_3
(A)	1	2	3	4
(B)	1	3	2	4
(C)	4	3	2	1
(D)	4	2	3	1



36. 1 mol of an ideal gas is subjected to following process



The magnitude of $q_{AB} - q_{AC}$ is : [Given : $T_1 = 300\text{ K}$, $T_2 = 400\text{ K}$, $R = 2\text{ cal mol}^{-1}\text{ K}^{-1}$]

- (A) 100 cal (B) 200 cal (C) 300 cal (D) 400 cal

Paragraph for Questions 37 - 38

To describe heat change for system maintained at a constant pressure, thermodynamic quantity is called as enthalpy (H). The enthalpy (H) of a system is its internal energy plus the product of its pressure and volume.

$$H = E + PV$$

Enthalpy is a state function. It is not influenced by the history of the system. The heat of reaction in a system at constant pressure equals its change in enthalpy.

$$\Delta H = \Delta E + P\Delta V = q_p$$

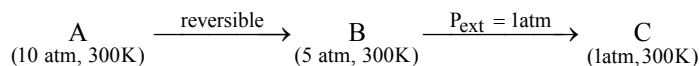
If we are dealing with a gas phase reaction, we can use the ideal gas law to perform these calculations : $P\Delta V = \Delta n_g RT$

The quantity Δn_g is the change in the number of moles of gas due to the reaction $\Delta n_g = n_{\text{gas, product}} - n_{\text{gas, reactant}}$.

37. The internal energy change for the reaction : $\text{NH}_4\text{Cl(aq)} + \text{NaNO}_2\text{(aq)} \longrightarrow \text{N}_2\text{(g)} + 2\text{H}_2\text{O(l)}$ is -333 kJ for each mole of N_2 produced. The formation of the mole of nitrogen expands the system 22.4 L at 1 atm pressure. What is the enthalpy change due to this reaction ?
 (A) -330.73 kJ (B) -333 kJ (C) 335.27 kJ (D) -348.6 kJ
38. Burning 0.1676 g of iron in a bomb calorimeter at 25°C produces 1.228 kJ heat : $4\text{Fe} + 3\text{O}_2 \longrightarrow 2\text{Fe}_2\text{O}_3\text{(s)}$ What is the enthalpy per mole of Fe_2O_3 for this reaction?
 (A) -819 kJ/mole (B) -823 kJ/mole (C) -846 kJ/mole (D) -912 kJ/mole

MULTIPLE CHOICE QUESTIONS

39. One mol of an ideal gas is expanded isothermally as following:



Which is/are correct statements. (Given : $\ln 4 = 1.4$, $R = 2\text{ cal mol}^{-1}\text{ K}^{-1}$)

- (A) Net work done by the system is -900 cal
 (B) Change in entropy for the system for path AB is 1.4 cal K^{-1} .
 (C) Change in free energy for path AB is -420 cal .
 (D) Change in entropy by the surrounding for path BC is -1.6 cal K^{-1}
40. Five moles of an ideal gas $\left(C_v = \frac{5}{2}R\right)$ are allowed to expand adiabatically and reversibly at 300 K from a pressure of 10 atmosphere to a pressure of 1 atmosphere . Then select correct statement(s) : ($10^{-2/7} = 0.51$)
 (A) The amount of work done is 3593.5 calories
 (B) The amount of work done is 15.033 kJ
 (C) The final temperature is 155.3 K
 (D) The final temperature is 100 K

41. If for an equation like $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ the ΔG (reaction) expression given by :

$$\Delta G = 4.73 \text{ kJ mol}^{-1} + 2.5 \text{ kJ mol}^{-1} \ln \left\{ \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}} \right\} \text{ then :}$$

- (A) if small amount of N_2O_4 is introduced in a closed container, then reaction moves in forward direction spontaneously.
 (B) Initially, ΔG° value is negative for the given decomposition.
 (C) At equilibrium, $Q_p = K_p = \text{anti log} \left| \frac{-\Delta G^\circ}{RT} \right|$
 (D) At equilibrium state ΔG must be zero.

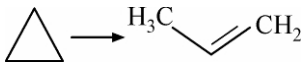
MATRIX MATCHING TYPE

42. Match the following

Column I

- (A) Heat of polymerization
 (B) Heat of neutralization
 (C) Heat of isomerization
 (D) Heat of combustion

Column II

- (P) $\text{NaOH} + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$
 (Q) 
 (R) $n\text{H}_2\text{C}=\text{CH}_2 \longrightarrow (-\text{CH}_2-\text{CH}_2-)_n$
 (S) $2\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
 (T) $\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$

43. Match the following

Column I

- (A) $2\text{O}_3(\text{g}) \longrightarrow 3\text{O}_2(\text{g})$
 (B) $3\text{O}_2(\text{g}) \longrightarrow 2\text{O}_3(\text{g})$
 (C) $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) + \text{Energy} \longrightarrow \text{A}_2\text{B}_2(\text{g})$
 (D) $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\ell)$ at room temperature, 25°C

Column II

- (P) $\Delta H = (+)\text{ve}$, $\Delta S = (-)\text{ve}$
 (Q) $\Delta H = (-)\text{ve}$, $\Delta S = (+)\text{ve}$
 (R) $\Delta G = (+)\text{ve}$
 (S) $\Delta H = (+)\text{ve}$, $\Delta S = (+)\text{ve}$
 (T) $\Delta G = (-)\text{ve}$

44. Match the following

Column I (Reaction)

- (A) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + x$
 (B) $\text{C}(\text{graphite}, \text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + x$
 (C) $\text{H}_2\text{O}_2(\ell) + x \longrightarrow 2\text{H}(\text{g}) + 2\text{O}(\text{g})$
 (D) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \longrightarrow \text{AgCl}(\text{s}) + x$

Column II

- (P) $\Delta H_{\text{precipitation}} = -x$
 (Q) $\Delta H_{\text{atomisation}} = x$
 (R) $\Delta H_{\text{combustion}} = -x$
 (S) $\Delta H_{\text{formation}} = -x$
 (T) Exothermic

45. Match the following

Column I (Different thermodynamic processes)	Column II (in P-V) (In P-V diagram in which P in Y-axis and V in X-axis corresponding slope for a mole of ideal gas)
(A) Isothermal process	(P) Infinity
(B) Adiabatic process	(Q) 0
(C) Isochoric process	(R) $-\gamma P/V$
(D) Isobaric process	(S) $-P/V$

SUBJECTIVE QUESTIONS46. If enthalpy of atomization of $\text{CH}_3\text{OH}(\ell)$ is ΔH (in kJ mol^{-1}). Calculate the value of z.

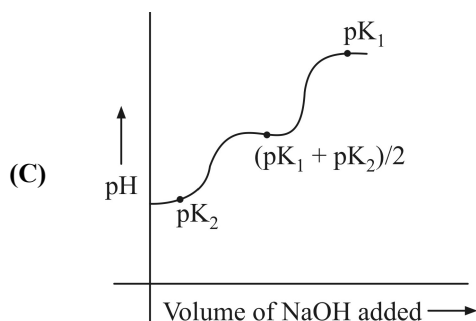
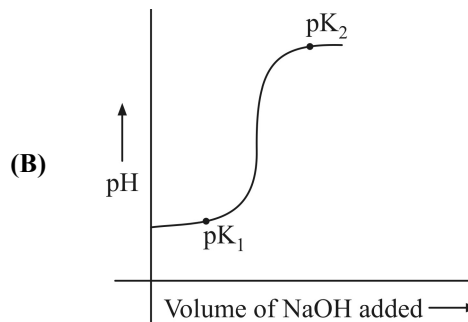
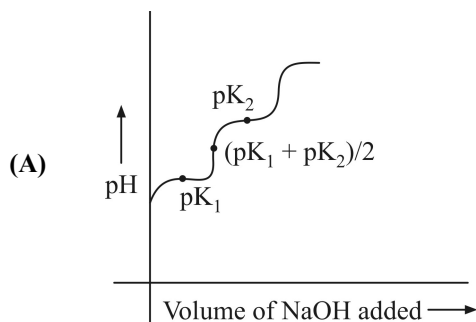
$$\text{Where } z = \frac{\Delta H - 60}{1000}$$

Given : $\Delta H_{\text{vap}}(\text{CH}_3\text{OH}(\ell)) = 28 \text{ kJ mol}^{-1}$.

Bond	C – H	C – O	O – H
Bond Energy (kJ mol^{-1})	414	326	464

47. Find molar entropy change for a monoatomic ideal gas when heated isochorically and irreversibly from 0°C to 273°C in cal?**CHEMICAL EQUILIBRIUM & IONIC EQUILIBRIUM****SINGLE CHOICE QUESTION**

48. One student makes a curve of a titration of a weak diprotic acid by NaOH of equivalent strength. Which of the following curves is correct regarding the above titrations :



(D) None of these

49. Which of the following is correct for the equilibrium of reaction $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g})$

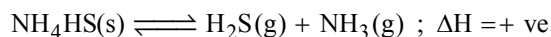
- (A) $P_{\text{H}_2} \propto P_{\text{H}_2\text{O}}$ (B) $P_{\text{H}_2} \propto \sqrt{P_{\text{H}_2\text{O}}}$ (C) $P_{\text{H}_2} \propto P_{\text{H}_2\text{O}}^2$ (D) $P_{\text{H}_2} \propto \frac{P_{\text{H}_2\text{O}}^2}{P_{\text{CO}}}$

50. Ammonium carbamate (X) dissociates as $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$
- Experiment – 1 :** In a closed vessel containing (X) in equilibrium, CO_2 is added such that partial pressure of CO_2 at new equilibrium equals to original total pressure.
- Experiment – 2 :** In a closed vessel containing (X) in equilibrium, NH_3 is added such that partial pressure of NH_3 at new equilibrium equals to original total pressure.
- Experiment – 3 :** Now CO_2 (having partial pressure equal to that of CO_2 at new equilibrium in Exp-2) and NH_3 (having partial pressure equal to that of NH_3 at new equilibrium in Exp – 1) are mixed together in another closed vessel containing some (X). Then the mass of (X) formed will be :
- (A) 1.1 g (B) 3.2 g (C) 6.3 g (D) No 'X' is formed
51. $\text{AO} \rightleftharpoons \frac{1}{2} \text{A}_2 + \frac{1}{2} \text{O}_2$; $K = 5 \times 10^5$ $\text{BO} \rightleftharpoons \frac{1}{2} \text{B}_2 + \frac{1}{2} \text{O}_2$; $K = 1.1 \times 10^{12}$
- $\text{CO} \rightleftharpoons \frac{1}{2} \text{C}_2 + \frac{1}{2} \text{O}_2$; $K = 2.3 \times 10^{18}$ $\text{DO} \rightleftharpoons \frac{1}{2} \text{D}_2 + \frac{1}{2} \text{O}_2$; $K = 1.4 \times 10^{21}$
- Which oxide is most stable ?
- (A) AO (B) BO (C) CO (D) DO
52. 100 ml 0.1 M H_3PO_4 is titrated with 150 ml 0.1 M NaOH. The pH of final solution is :
- (Given : $K_1 = 10^{-3}$, $K_2 = 10^{-8}$, $K_3 = 10^{-12}$)
- (A) 3 (B) 8 (C) 12 (D) 10
53. The range of a basic indicator InOH is 3.4 to 4.6. The minimum ratio of $\frac{[\text{In}^+]}{[\text{InOH}]}$ so that the solution will give colour change in In^+
- (A) 10 (B) 8 (C) 6 (D) 4
54. We have taken A(g) in closed container. The following reactions take place
- $$2\text{A}(\text{g}) \rightleftharpoons \text{A}_2(\text{g}) ; \quad \text{A}_2(\text{g}) \rightleftharpoons \text{B}(\text{g}) + \text{C}(\text{s})$$
- At a particular instant, degree of association of A is 0.5 and degree of decomposition of A_2 is also 0.5. The vapour density of A and equilibrium mixture are D and d respectively. Select the correct option.
- (A) $\frac{D}{d} = \frac{6M_A + M_B}{6M_A}$ (B) $\frac{D}{d} = \frac{6M_A - M_B}{6M_A}$ (C) $\frac{D}{d} = \frac{6M_A}{6M_A - M_B}$ (D) $\frac{D}{d} = \frac{6M_A}{6M_A + M_B}$
55. 10 ml of 0.1 M HCl solution is added in 90 ml of a buffer solution having 0.1 M NH_4OH and 0.1 M NH_4Cl . The percentage change in pH of solution is (K_a of $\text{NH}_4^+ = 5 \times 10^{-10}$, $\log 2 = 0.3$)
- (A) $\frac{10}{4.7}$ % increase (B) $\frac{10}{4.7}$ % decrease (C) $\frac{10}{9.3}$ % increase (D) $\frac{10}{9.3}$ % decrease

MULTIPLE CHOICE QUESTIONS

56. Consider the equilibria :
- $\text{Hb}(\text{aq}) + 4\text{CO}(\text{g}) \rightleftharpoons \text{Hb}(\text{CO})_4(\text{aq})$ (deep red) K_{eq1}
 - $\text{Hb}(\text{aq}) + 4\text{O}_2(\text{g}) \rightleftharpoons \text{Hb}(\text{O}_2)_4(\text{aq})$ (light red) K_{eq2}
- (Hb = Hemoglobin in blood)
- Identify the correct statement(s)
- (A) On moving from lower to higher altitudes, at higher altitudes, equilibria 2 shifts towards right
- (B) At plain area equation $K_{\text{eq1}} < K_{\text{eq2}}$ (C) At hilly area $K_{\text{eq1}} > K_{\text{eq2}}$
- (D) People living on hills have more hemoglobin in their body than people living on plains.

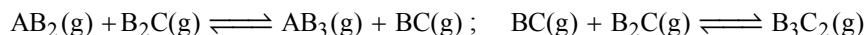
57. The dissociation of ammonium carbamate may be represented by the equation.



The equilibrium will be shifted from left to right if there is :

- (A) A decrease in pressure (B) An increase in temperature
(C) An increase in amount of $\text{NH}_4\text{HS(s)}$ (D) Addition of inert gas at constant pressure

58. If two gases AB_2 and B_2C are mixed the following equilibrium are established.



If the reaction is started with AB_2 with B_2C then identify the false statement at equilibrium

- (A) $[\text{AB}_3] = [\text{BC}]$ (B) $[\text{AB}_2] = [\text{B}_2\text{C}]$ (C) $[\text{AB}_3] < [\text{B}_3\text{C}_2]$ (D) $[\text{AB}_3] > [\text{BC}]$

59. A solution contains 0.01 M conc. each of Zn^{2+} , Mg^{2+} and Mn^{2+} ions. They are to be precipitated as their sulphide by passing H_2S gas into the solution and the solution is saturated with 0.1 M H_2S .

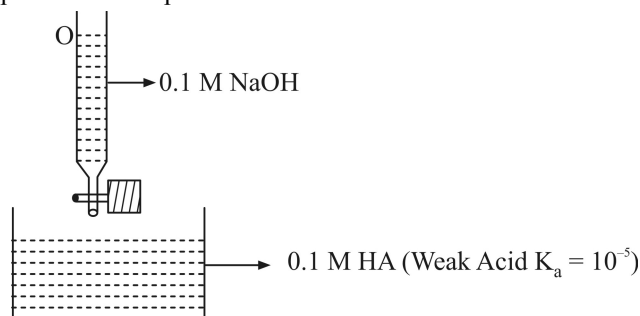
[Given : $K_{sp}(\text{ZnS}) = 10^{-18}$, $K_{sp}(\text{MnS}) = 10^{-22}$, $K_{sp}(\text{MgS}) = 10^{-12}$, $K_a(\text{H}_2\text{S}) = 10^{-21}$]

Identify which is/are correct statement

- (A) Only MnS is precipitated in pH range of 1 to 3
(B) ZnS is precipitated when the pH > 3
(C) MgS gets precipitated when pH > 6.
(D) MgS get precipitated when pH < 3.

MATRIX MATCHING TYPE

60. Consider the following experimental setup :



Column I (Expression for pH)

Column II (Extent of titration)

- | | |
|--|----------------------|
| (A) $\text{pK}_a + \log \frac{[\text{salt}]}{[\text{Acid}]}$ | (P) No titration |
| (B) $\frac{1}{2}(\text{pK}_a - \log C)$ | (Q) Titration < 100% |
| (C) $14 + \log[\text{OH}^-]$ | (R) Titration = 100% |
| (D) $\frac{1}{2}(\text{pK}_w + \text{pK}_a + \log C)$ | (S) Titration > 100% |

61. Match the column :

Column I

Column II

- | | | |
|--|------------------------------------|--|
| (A) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{NO(g)}$ | Pressure is increased | (P) Rate of forward reaction increases |
| (B) $\text{SO}_2\text{(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightleftharpoons \text{SO}_3\text{(g)}$ | Volume is halved | (Q) Rate of backward reaction increases |
| (C) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$ | $\text{NH}_3\text{(g)}$ is added | (R) Rate of forward reaction = Rate of backward reaction |
| (D) $\text{PCl}_5\text{(g)} \rightleftharpoons \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$ | Inert gas added at constant volume | (S) Reaction Quotient $\neq$ Equilibrium constant |

62. Match the column

Column I	Column II
(A) Solubility of AgCl in water	(P) 10^{-5} M
(B) Solubility of Ag_2SO_4 in water	(Q) 1.435×10^{-6} gm/ml
(C) Solubility of Ag_2SO_4 in 0.1 M AgNO_3	(R) 4×10^{-13} M
(D) Solubility of Ag_2SO_4 in 0.1 M Na_2SO_4	(S) 10^{-7} M

Given : $K_{sp}(\text{AgCl}) = 10^{-10}$; $K_{sp}(\text{Ag}_2\text{SO}_4) = 4 \times 10^{-15}$

63. Match the column

Column I	Column II
(A) 1L, 1M Na_2CO_3 + 1L, 1M H_2CO_3 ; [$K_{a1}(\text{H}_2\text{CO}_3) = 10^{-6}$, $K_{a2} = 10^{-9}$]	(P) pH = 7.5
(B) 1L, 0.2 M $(\text{CH}_3\text{COO})_2\text{Ca}$ + 1L, 0.2M HCl solution [$K_a(\text{CH}_3\text{COOH}) = 10^{-5}$]	(Q) pH = 1
(C) 1L, 0.2 M NH_4Cl + 1L, 0.2 M HCl solution	(R) pH = 9
(D) 1L, 0.2 M NH_4Cl + 1L, 0.1 M NaOH solution [$K_b(\text{NH}_4\text{OH}) = 10^{-5}$]	(S) pH = 5

SUBJECTIVE QUESTIONS

64. $x \times 10^{-1}$ moles of NH_3 is added to a 25×10^{-3} M Ag^+ solution to just prevent the precipitation of AgCl when $[\text{Cl}^-]$ reaches to 10^{-3} M. What is the value of x? Assume total volume of solution is 1 L.
[Given : $K_{sp}(\text{AgCl}) = 10^{-10}$, $K_{\text{instability}}[\text{Ag}(\text{NH}_3)_2]^+ = 10^{-8}$]
65. A solid substance "X" decompose into two gaseous products "Y" and "Z" as $\text{X(s)} \rightleftharpoons 2\text{Y(g)} + \text{Z(g)}$
If at equilibrium, some Z(g) at 1 atm is added at constant volume condition, 10% of Y(g) solidified before the equilibrium was re-established. Determine total pressure (in bar) at final equilibrium.
66. A compound "A" in aqueous phase dissociates to form unknown number of moles of gaseous products. Find the number of moles of gaseous products formed per mole of "A" reacted, if K_p of the reaction is 8 and $K_c = 2$ at temperature of $\left(\frac{2}{R}\right)$ Kelvin [R = gas constant]
67. Find out negative logarithm of solubility of solid Zn(OH)_2 at 25°C and pH = 6.
Given : $\text{Zn(OH)}_2(\text{s}) + \text{aq} \rightleftharpoons \text{Zn(OH)}_2(\text{aq})$, $K_1 = 10^{-6}$
 $\text{Zn(OH)}_2(\text{aq}) \rightleftharpoons \text{Zn(OH)}^+(\text{aq}) + \text{OH}^-(\text{aq})$, $K_2 = 10^{-7}$
 $\text{Zn(OH)}^+(\text{aq}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{OH}^-(\text{aq})$, $K_3 = 10^{-4}$
 $\text{Zn(OH)}_2(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{Zn(OH)}_3^-(\text{aq})$, $K_4 = 10^3$
 $\text{Zn(OH)}_3^-(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{Zn(OH)}_4^{2-}(\text{aq})$, $K_5 = 10$
68. Calculate the pH of 0.005 M $(\text{CH}_3\text{COO})_2\text{Ba}$. [Given : $K_a(\text{CH}_3\text{COOH}) = 10^{-6}$]
69. Given $\text{H}_2(\text{g}) + \text{S(s)} \rightleftharpoons \text{H}_2\text{S(g)}$
 $\Delta G_f^\circ(\text{H}_2\text{S, gas}) = -6.909 \text{ kcal}$
If K_p for this reaction is 1×10^x at 300 K, calculate the value of x.

70. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 300 K

Given	Gas	NH_3	N_2	H_2
	$\Delta_f H^\circ(\text{J})$	-45×10^3	0	0
	$\Delta_f S^\circ(\text{JK}^{-1})$	+ 200	+ 190	130

Calculate the value of Z

Where $z = \left| \frac{(\Delta_f G^\circ \text{ in kJ/mole})}{2} \right|$

71. Calculate the percent dissociation of $\text{H}_2\text{S}(\text{g})$ if 0.1 mole of H_2S is kept in 0.4 litre vessel at 1000 K.

For the following reaction: $2\text{H}_2\text{S}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{S}_2(\text{g})$; $K_c = 1.0 \times 10^{-6}$

ANSWERS KEY

Advanced Chemistry Assignment-1A | Illuminati-2019

Physical Chemistry Class XIth

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	B	A	D	B	C	B	A	B	D	C	C	A	B	ABCD
16	17	18	19	20	21	22					23			
ABC	BCD	ABCD	ABCD	ACD	C	A - P ; B -Q R ; C - PQ ; D - RS					A -S ; B -R ; C - P ; D - Q			
24					25	26	27	28	29	30	31	32	33	34
A - R ; B - R ; C - S ; D - P					6	9	6	5	2	4	5	4	B	A
35	36	37	38	39		40	41	42						
D	A	A	B	ABCD		ABC	ACD	A - R ; B - P, S ; C - Q ; D - T						
43				44					45				46	
A - QT ; B - P R ; C - PR ; D - TS				A - R T ; B - R S T ; C - Q ; D - P T					A - S ; B - R ; C - P ; D - Q				2	
47	48	49	50	51	52	53	54	55	56	57	58		59	
2	A	B	D	A	B	D	D	D	CD	ABD	ABC		ABC	
60				61					62					
A - Q ; B - P ; C - PQRS ; D - R				A - R ; B - P, S ; C - QS ; D - R					A - PQ ; B - P ; C - R ; D - S					
63				64	65	66	67	68	69	70	71			
A - P ; B - S ; C - Q ; D - R				1	9	2	1	9	5	9	2			